Notice that the remaining market impact cost and alpha trend are the only components that can be affected by the trading strategy. Investors will seek to manage projected costs by trading either faster or slower based on market conditions, price momentum, and desired adaptation tactic.

The timing risk of these unexecuted shares represents the uncertainty surrounding the market impact estimate for the remaining shares. This is:

$$E_t[TR'] = \sigma \cdot \sqrt{(1-\theta) \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha_t}} \cdot P_0 \tag{8.31}$$

The projected cost of the order is then,

$$\begin{aligned} E_t\left[Projected\ Cost\left(\$/share\right)\right] = Realized(Cost(t)) = + E_t[Momentum] \\ + E_t[MI'] + E_t[\text{Alph}a'] \end{aligned} \tag{8.32}$$

Written formulaically this is:

$$\begin{aligned} E_t\left[Projected\right] = \theta \cdot (\overline{P}_t - P_0) \cdot Side \cdot (1-\theta)(((\overline{P}_t - P_0) \cdot Side) \\ + (b_1 \cdot I' \cdot \alpha_t + (1-b_1) \cdot I')) + (1-\theta) \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha_t} \cdot \mu_t \end{aligned} \tag{8.33}$$

Notice that this projected cost expression consists of a component that is "sunk" and "unavoidable" and a component that is "controllable." The sunk cost component is comprised of the realized cost for the transacted shares. The unavoidable component is comprised of the momentum cost and permanent market impact cost. The controllable component is comprised of the price impact of the shares that are to be traded and alpha cost of these shares. This is also the component that can be managed through proper selection of the trading strategy.

Recall that our initial cost is denoted as $E_0[Cost]$ and is expressed in \$/share but could also be expressed in total dollars or in basis points.

For simplicity, we proceed in the examples below without the alpha term in basis point units. We leave it as an exercise for our readers to work through the math including the alpha cost component.

Let K_t represent the costs that are unavoidable (realized, momentum, and permanent):

$$K_t = \theta \cdot (\overline{P}_t - P_0) \cdot Side + (1-\theta) \cdot ((\overline{P}_t - P_0) \cdot Side) + (1-\theta) \cdot (1-b_1) \cdot I' \tag{8.34}$$

Then the projected cost can be simplified as follows:

$$E_t(Cost) = K_t + (1-\theta)(b_1 \cdot I' \cdot \alpha_t)$$